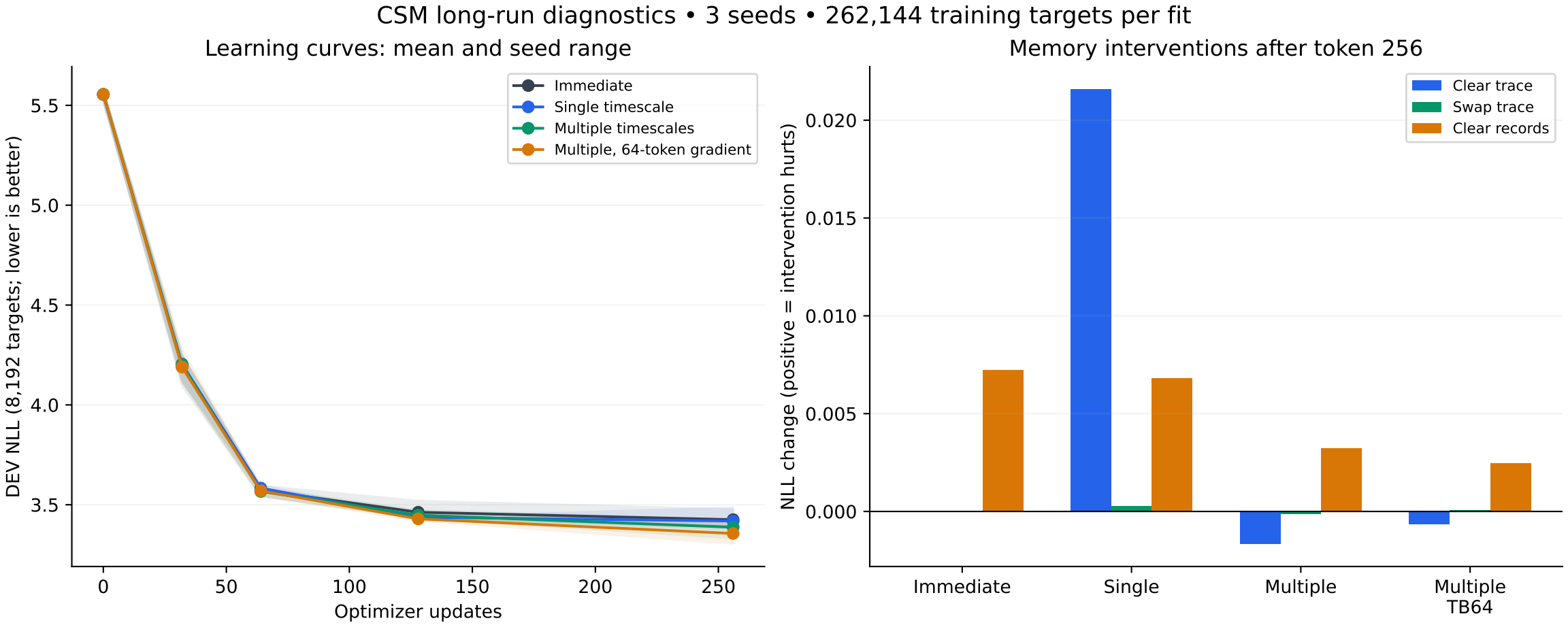
CSM long-run diagnostics • 3 seeds • 262,144 training targets per fit
Learning curves: mean and seed range
Immediate
Single timescale
Multiple timescales
Multiple, 64-token gradient
DEV NLL (8,192 targets; lower is better)
5.5
5.0
4.5
4.0
3.5
0
50
100
150
200
250
Optimizer updates
Memory interventions after token 256
Clear trace
Swap trace
Clear records
NLL change (positive = intervention hurts)
0.020
0.015
0.010
0.005
0.000
Immediate
Single
Multiple
Multiple TB64